

US02CBCA01 || Advanced C and Introduction to Data Structure

Total Marks: 70

[10]

- [P.T.O.]**

Q.2 Explain following in brief. (Attempt any ten)

[20]

- 1 Explain malloc() function.
- 2 List out benefits of pointers.
- 3 Differentiate between '*' and '&' operators in pointers.
- 4 List file modes available to manage the file in C.
- 5 Explain typedef in brief with suitable example.
- 6 Differentiate: structure and union
- 7 Write basic operation performed on stack.
- 8 State various Applications of Data structure.
- 9 Draw the Hierarchical Structure of Data Structure.
- 10 Define: Circular Queue and Priority Queue.
- 11 State various Applications of Linked List.
- 12 Define: doubly linked list

Q.3 A. What is dynamic memory allocation? Explain realloc() and free() function.

[06]

B. Define pointer variable. How can we declare and initialize pointer variable? Explain with example.

[04]

OR

Q.3 A. Explain pointer arithmetic with example.

[05]

B. Explain pointer and array with syntax and example.

[05]

Q.4 A. What is structure? Explain its declaration and assigning values to members of structure.

[05]

B. Explain fopen() and fclose() function with example.

[05]

OR

Q.4 A. Explain array of structures with syntax and suitable examples.

[05]

B. Explain structure within structure with syntax and example.

[05]

Q.5 A. Write an algorithm to insert and delete an element from a Stack.

[06]

B. Write a short note on primitive and non-primitive data structure.

[04]

OR

Q.5 A. Write an algorithm to change and access an element from a Stack.

[06]

B List and Explain operations performed on Data Structure.

[04]

Q.6 A What is queue? Write an algorithm to Insert and Delete an element into a simple queue.

[10]

OR

Q.6 A What is linked list? Write an algorithm to insert an element at the beginning of a singly linked list and delete an element from singly linked list.

[10]

Best of Luck,

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